

StatSkill AI - Ministry & Department Competency Framework

StatSkill AI — Ministry & Department-Specific Competency Framework

6 Domains × Role Grade × Ministry/Department posting

1. The 6 Domains

#	Domain	Nature
D1	Statistical (Survey Design, Sampling, National Accounts, Price Stats, Labour Stats, Agri Stats, Industrial Stats, SDG Indicators, Metadata Standards, Data Quality Frameworks)	Common — every officer needs the <i>general method</i>
D2	Technical (Python, R, SQL, Stata, SPSS, SAS, GIS, Data Viz, AI/ML, Cloud, APIs, Open Data)	Common
D3	Digital Governance (Cybersecurity, Data Privacy, Digital Signatures, Govt. Cloud, DPI)	Common
D4	Behavioural (Communication, Ethics, Decision Making)	Common
D5	Managerial (Leadership, Project Management, Change Management)	Common
D6	Sectoral/Domain-Specific Knowledge	Ministry-specific — this is what makes each ministry/department’s framework unique

D1–D5 use the same 6-role structure everywhere. **D6 is where each ministry differs**, and it also drives what the officer is *called* — a JSO doing water-body enumeration is doing genuinely different work from a JSO doing school data validation, even though both are “R1” on the grade ladder.

2. Designation-Naming Rule (how titles vary by ministry/department)

Every posting has a **Grade** (R1–R6, fixed ladder) and a **Sector Tag** (derived from the department). The displayed designation = Grade-title pattern + Sector Tag.

Grade	Base title pattern
R1	[Sector] Field Enumerator / Data Collector
R2	[Sector] Statistical Supervisor
R3	Assistant Director ([Sector] Statistics)
R4	Deputy Director ([Sector] Statistics)
R5	Director ([Sector] Statistics)
R6	Additional DG / Statistical Adviser ([Sector])

Example: in Jal Shakti’s Minor Irrigation Directorate, R3 = “**Assistant Director (Water Resources Statistics)**”; in Education’s UDISE+ cell, R3 = “**Assistant Director (Education Data Systems)**”. Same grade, same D1–D5 requirement, different D6 + different title.

3. Proficiency Generation Rule (applies to D6 for every department)

Whatever the sectoral competencies are, apply this same shape (matches the pattern already used for D1 in the earlier framework):

Grade	Sectoral (D6) proficiency
R1	L1 (awareness of sector-specific forms/concepts)
R2	L2 (independent field-level application)
R3	L3 (analysis & tabulation in that sector)
R4	L4 (methodology ownership for that sector)
R5	L4 (approves sector methodology/quality)
R6	L2–L3 (policy-level oversight, not hands-on)

This means you don’t need to hand-design every proficiency cell — only the **competency names per department** change; the level pattern is reusable.

4. Ministry × Department × Sector Tag × D6 Competencies

(from your registration dropdowns — designation examples shown at R3; apply the Section 2 formula for R1/R2/R4/R5/R6)

Ministry of Jal Shakti

Department	Sector Tag	D6 Competencies	R3 designation
Minor Irrigation Census & Water Bodies Directorate	Water Resources	Minor Irrigation Census Methodology, Water Body Enumeration Standards	Asst. Director (Water Resources Statistics)
National Minor Irrigation Census & Groundwater Data Division	Groundwater Data	Groundwater Level Monitoring, Aquifer Data Classification	Asst. Director (Groundwater Statistics)
Census of Water Bodies & Geo-Tagging Survey Unit	Geo-Water Survey	GPS/Geo-tagging Protocols, Water Body Inventory Standards	Asst. Director (Geo-Water Survey)

Central Water Commission (CWC) — Hydrological & Water Statistics	Hydrology	Hydrological Data Analysis, River/Reservoir Flow Statistics	Asst. Director (Hydrological Statistics)
--	-----------	---	--

Ministry of Education

Department	Sector Tag	D6 Competencies	R3 designation
Statistics & Educational Survey Division	Education	School Survey Methodology, Enrollment/Dropout Indicators	Asst. Director (Education Statistics)
AISHE Data Analytics Cell	Higher Education	Higher-Ed Survey Standards (AISHE), Institution-level Data Validation	Asst. Director (Higher Education Data)
UDISE+ Data Directorate	School Data Systems	UDISE+ Schema & Codes, School Infrastructure Indicators	Asst. Director (Education Data Systems)
PGI & NAS Unit	Learning Outcomes	Performance Grading Index Methodology, Learning Assessment Statistics	Asst. Director (Learning Outcomes Statistics)

Ministry of Rural Development

Department	Sector Tag	D6 Competencies	R3 designation
Statistics, Monitoring & Evaluation Division	Rural Dev	Scheme M&E Indicators, Rural Development Survey Design	Asst. Director (Rural Dev Statistics)
DISHA/SECC Analytics	Socio-Economic Census	SECC Data Structure, Deprivation Indicator Computation	Asst. Director (Socio-Economic Census)
MGNREGA Real-time MIS	Employment Guarantee	MIS Wage/Workday Metrics, Real-time Scheme Dashboards	Asst. Director (MGNREGA Analytics)
PMGSY GIS Directorate	Rural Connectivity	Road Connectivity GIS Mapping, Infrastructure Progress Metrics	Asst. Director (Rural Connectivity GIS)

Ministry of Labour & Employment

Department	Sector Tag	D6 Competencies	R3 designation
Labour Bureau	Labour Statistics	Labour Force Survey Concepts, Employment Classification Standards	Asst. Director (Labour Statistics)
CPI-IW Division	Industrial Price Index	CPI-IW Basket & Weight Methodology	Asst. Director (CPI-IW)
CPI-AL/RL	Agri-Labour Price Index	Rural Wage Index Construction	Asst. Director (CPI-AL/RL)
Employment-Unemployment/Wage Cell	Employment Survey	Wage Rate Survey Design, Unemployment Rate Estimation	Asst. Director (Employment Statistics)
DGE	Employment Services	Employment Exchange Data Standards	Asst. Director (Employment Data)
NCS Analytics	Labour Market Info	Labour Market Information Systems (LMIS)	Asst. Director (Labour Market Analytics)

Ministry of Health & Family Welfare

Department	Sector Tag	D6 Competencies	R3 designation
Statistics & Data Analytics Division	Health Statistics	Vital Statistics Methodology, Health Indicator Frameworks	Asst. Director (Health Statistics)
NFHS & Demographic Cell	Demography	Demographic Survey Design (NFHS), Fertility/Mortality Indicators	Asst. Director (Demographic Statistics)
CBHI	Health Records	Health Records Compilation Standards	Asst. Director (Health Records)
HMIS	Digital Health	HMIS Data Schema, Facility-level Health Reporting	Asst. Director (Digital Health Analytics)

Ministry of Agriculture & Farmers Welfare

Department	Sector Tag	D6 Competencies	R3 designation
DES-Agriculture	Agri Statistics	Crop Estimation Survey Methodology, Agri-Economic Indicators	Asst. Director (Agriculture Statistics)
Crop Estimation/Advance Estimates Wing	Crop Statistics	Crop Cutting Experiments, Advance Estimate Methodology	Asst. Director (Crop Statistics)
Agricultural Census & Land Use Division	Land Use Survey	Agricultural Census Concepts, Land Use Classification	Asst. Director (Land Use Statistics)
ISAS Directorate	Integrated Agri Stats	ISAS Scheme Data Standards	Asst. Director (Integrated Agri Statistics)
MNCFC	Remote Sensing/GIS	Satellite-based Crop Forecasting, GIS for Agriculture	Asst. Director (Agri Remote Sensing)

Ministry of Finance

Department	Sector Tag	D6 Competencies	R3 designation
Economic Division	Macro-Economics	Macroeconomic Forecasting Models, Economic Survey Compilation	Asst. Director (Economic Statistics)
Budget Division — Fiscal Statistics	Fiscal Statistics	Fiscal/Revenue Data Classification	Asst. Director (Fiscal Statistics)
CBDT Tax Analytics Cell	Direct Tax Statistics	Direct Tax Data Analysis	Asst. Director (Tax Statistics)
CBIC GST Analytics	Indirect Tax Statistics	GST/Trade Data Intelligence	Asst. Director (GST Analytics)
Financial Services — Banking Stats	Financial Inclusion	Banking/Credit Flow Indicators	Asst. Director (Financial Statistics)

Ministry of Statistics & Programme Implementation (MoSPI)

Department	Sector Tag	D6 Competencies	R3 designation
SDRD (Survey Design)	Survey Methodology	Advanced Sample Design, Survey Instrument Development	Asst. Director (Survey Design)
FOD	Field Operations	Field Data Collection Management, CAPI Systems	Asst. Director (Field Operations)
DPD	Data Processing	Large-scale Data Processing Pipelines	Asst. Director (Data Processing)
NAD	National Accounts	GDP/GVA Compilation, Base-year Revision Methods	Asst. Director (National Accounts)
ESD	Industrial/Economic Stats	ASI/IIP Concepts, Business Register Maintenance	Asst. Director (Economic Statistics)
PSD	Price Statistics	CPI Construction Methodology	Asst. Director (Price Statistics)
SSD	Social Statistics	SDG Indicator Compilation, Social Survey Concepts	Asst. Director (Social Statistics)

This is a genuinely strong demo point: show 2 full worked examples live, then show the same engine instantly generating the other 42 departments' frameworks from the Section 4 data table — proves the system is data-driven, not hardcoded, which is exactly what judges want to see for a scalable GovTech tool.